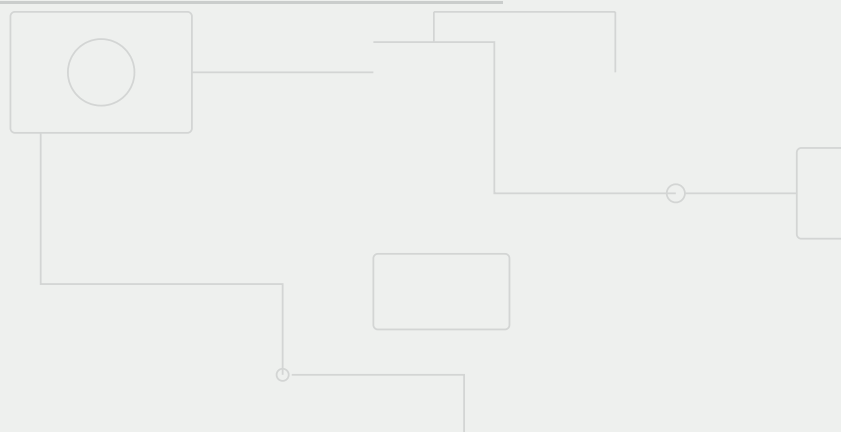


PUMP MOTOR CONTROL SYSTEMS

Never let a dry run burn your motor.

Three control panels for pump protection and automation — manual, tank-to-tank automatic, and timer-based borewell control — each also available with AquaControl app connectivity. Overload, short-circuit and dry-run protection is built into every panel.

01	Manual Pump Control & Protection Panel	SINGLE / THREE-PHASE
02	Tank-to-Tank Automatic Controller	AUTO / MANUAL
03	Borewell Pump Control & Protection Panel	TIMER-BASED AUTOMATIC
04	Manual Pump Control & Protection Panel	APP CONTROLLED
05	Tank-to-Tank Automatic Controller	APP CONTROLLED
06	Borewell Pump Control & Protection Panel	APP CONTROLLED



Manual Pump Control & Protection Panel

COMPLETE MOTOR PROTECTION · SIMPLE CONTROL · RELIABLE OPERATION

A robust and easy-to-use manual pump control panel designed for safe operation and protection of water pumps and motors. The operator starts and stops the pump directly from the panel, while the panel continuously monitors electrical parameters and protects the motor against abnormal operating conditions.

- **Manual Start / Stop** — dedicated controls let the operator start and stop the pump directly from the panel.
- **Complete pump protection** — guards against the common electrical and operating faults that damage motors.
- **Overload protection** — trips the motor when current exceeds the configured operating limit.
- **Short-circuit protection** — protects wiring and equipment against short-circuit conditions.
- **Dry-run protection** (optional) — current-based detection of low-load operation from insufficient water or lost pump load.
- **Digital monitoring** (optional) — supply voltage, motor current, fault status and protection settings on the display.
- **Fault indication** — individual indication for overload, dry run and general fault.



FRONT PANEL CONTROLS

MAIN ON / OFF Isolates the complete control panel from the incoming supply.

START Starts the pump when all protection conditions are healthy.

STOP Immediately stops the pump.

DIGITAL DISPLAY Shows current and fault information, depending on the selected model.

Protection logic, variants & specifications

HOW THE PROTECTION SYSTEM RESPONDS

Supply Healthy → Protection Healthy → START → Pump ON
Fault Detected → Pump OFF → Fault Indication

AVAILABLE PROTECTIONS

Short Circuit	Protects wiring and equipment.
Overload	Protects the motor against excessive current.
Dry Run	Detects low-load / no-load pump operation.

SINGLE-PHASE & THREE-PHASE VARIANTS

SINGLE-PHASE MODEL

Suitable for

- Domestic water pumps
- Submersible pumps
- Monoblock pumps
- Small transfer pumps
- Agricultural pumps

THREE-PHASE MODEL

Suitable for

- Submersible & open-well pumps
- Monoblock & industrial pumps
- Agricultural pumps
- Water transfer systems
- Commonly paired with DOL or other starters

TECHNICAL SPECIFICATIONS

OPERATION	Manual
SENSOR REQUIREMENT	Not required
SUPPLY	230 VAC / 415 VAC, 50 Hz *
PUMP TYPE	Single Phase / Three Phase *
STARTER	DOL
MOTOR PROTECTION	Overload · Dry Run (current-based) *
ENCLOSURE / MOUNTING	MS Powder Coated · Wall Mount

Applications: water transfer & distribution, irrigation, borewell & open-well pumps, sump pumps, process-water & industrial transfer, maintenance and testing. *Specification depends on the selected model.

Tank-to-Tank Automatic Controller

SMART WATER TRANSFER · AUTOMATIC LEVEL CONTROL · PUMP PROTECTION

Automates the transfer of water between a source tank and a destination tank with intelligent level-based pump control. The controller continuously monitors water levels in both tanks and automatically operates the transfer pump according to the configured minimum and maximum levels. No manual monitoring. No unnecessary pump operation. No tank overflow.

- **Fully automatic operation** — starts and stops the transfer pump according to the destination tank's water level.
- **Dual tank monitoring** — watches both the source and destination tank for safe, reliable transfer.
- **Dry-run protection** — prevents the pump running when the source tank has insufficient water.
- **Overflow protection** — stops the pump once the destination tank reaches its configured maximum level.
- **Auto / Manual mode** — automatic by tank level, or user-controlled for maintenance and testing.
- **Intelligent control** — microcontroller-based operation for accurate, repeatable tank-level management.



OPERATING MODES

AUTO MODE

Fully automatic operation based on source and destination tank levels.

MANUAL MODE

Operator starts/stops the pump directly — for testing, maintenance or emergency operation.

How it works & specifications

HOW IT WORKS

- 1

Source Tank
The controller checks whether sufficient water is available in the source tank.
- 2

Destination Tank
The controller continuously monitors the water level of the destination tank.
- 3

Pump Start
When the destination tank reaches its LOW level and the source tank has enough water, the transfer pump starts.
- 4

Automatic Filling
The pump continues transferring water while the controller monitors both tanks.
- 5

Pump Stop
When the destination tank reaches its FULL level, the controller automatically switches the pump OFF.
- 6

Source Protection
If the source tank reaches its minimum level during operation, the controller stops the pump to prevent dry running.

LEVEL CONTROL

Source • LOW	Minimum safe water level.
Source • FULL	Source tank has adequate water.
Destination • LOW	Pump start condition.
Destination • FULL	Pump stop condition.

TECHNICAL SPECIFICATIONS

CONTROLLER TYPE	Microcontroller-based, tank-to-tank
TANK MONITORING	Source + Destination
SENSOR INPUT	Water-level sensors
PUMP CONTROL	Relay output
SUPPLY / PUMP TYPE	230 VAC, 50 Hz • Single / Three Phase *
PROTECTION	Dry Run • Overflow • Overload • Short Circuit *
INDICATION / ENCLOSURE	7-Segment Display * • Metal, Wall Mount

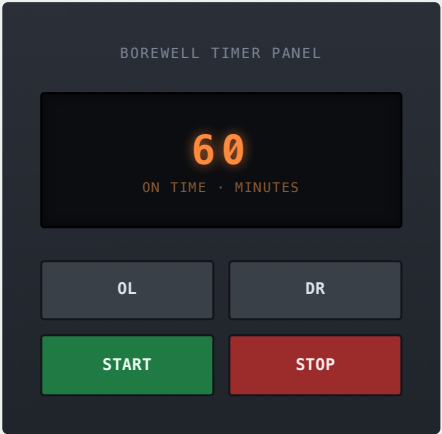
Applications: residential buildings, apartments, villas & commercial buildings, hotels, hospitals, schools, farms, underground-to-overhead tank filling, borewell/sump to overhead tank. *Specification depends on the selected model.

Borewell Pump Control & Protection Panel

TIMER-BASED AUTOMATIC OPERATION · COMPLETE PUMP PROTECTION

A reliable automatic control panel for borewell and submersible pumps, providing scheduled ON/OFF operation with comprehensive electrical and motor protection. The pump runs on a programmed ON TIME / OFF TIME cycle, with an upper-tank level sensor completing the control loop.

- **Timer-based automatic operation** — set the pump's ON TIME and OFF TIME; the controller runs the programmed cycle automatically.
- **Upper tank sensor included** — works alongside the programmed timer for reliable, predictable filling.
- **Complete pump protection** — overload, short-circuit and dry-run / low-current protection, depending on model and sensor configuration.
- **Manual mode** — direct START / STOP control for maintenance, testing, commissioning and emergency operation.



TIMER CONTROL

ON TIME = 60 minutes → the pump runs for 60 minutes
OFF TIME = 120 minutes → the pump waits 120 minutes before the next cycle
Automatic cycle: ON → OFF → ON → OFF – repeats until stopped

Dry-run logic, protection & specifications

DRY-RUN PROTECTION (CURRENT-BASED)

Low Current Detected → Dry Run → Pump OFF → Fault Indication

If the borewell runs dry or the pump load drops below the configured threshold, the panel cuts power — helping protect the submersible pump when sufficient borewell water is unavailable.

ELECTRICAL PROTECTION

Short Circuit	Helps protect the panel, wiring and motor circuit against short-circuit conditions.
Overload	Electronic or thermal overload protection disconnects the motor when current exceeds the configured limit.

BOREWELL-SPECIFIC FEATURES

TIMER + SENSOR CONTROL	POWER-FAILURE RECOVERY
The programmed ON/OFF cycle works together with the upper-tank sensor.	After an interruption, resumes operation per the configured mode.

TECHNICAL SPECIFICATIONS

APPLICATION	Borewell / Submersible Pump
CONTROL METHOD	Timer-Based
UPPER TANK SENSOR	Required
OPERATING MODES	Auto / Manual
DRY RUN	Current-based *
SUPPLY / PUMP TYPE	Single / Three Phase * · Submersible
ENCLOSURE	Wall Mount

Applications: borewell submersible pumps, farm & agricultural irrigation, residential borewell systems, underground-to-overhead tank filling, apartment & commercial water systems, remote pumping locations. *Specification depends on the selected model.

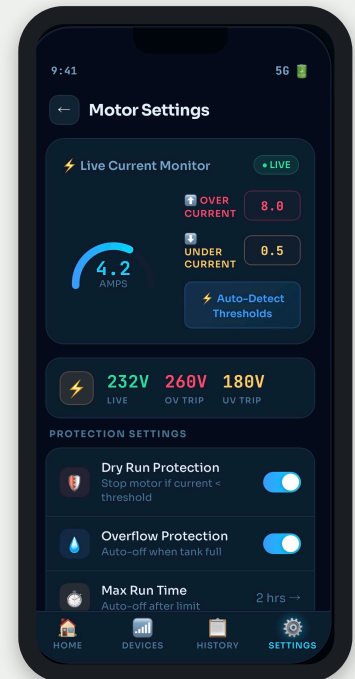
Manual Pump Control & Protection Panel — App Controlled

SAME PROTECTION · REMOTE CONTROL · LIVE MONITORING

+ AQUACONTROL APP

Everything in the Manual Pump Control & Protection Panel (Product 01), connected to the AquaControl app — so the pump can be started, stopped and monitored from a phone, not only from the panel itself.

- **Remote Start / Stop** — turn the pump ON or OFF from the app, from anywhere.
- **Live current & voltage monitoring** — see actual amps and volts on your phone, refreshed on demand.
- **Configurable protection thresholds** — set over-current and under-current limits from the app, or use Auto-Detect Thresholds to learn them from the running pump.
- **Offline alert** — notified the moment the panel stops responding, not just when you happen to check.
- **Fault visibility** — overload, dry-run and fault status visible remotely, in addition to the panel's own indication.



Protection logic, single/three-phase variants and technical specifications are unchanged from Product 01 (page 03) — the app adds remote visibility and control on top of the same panel.

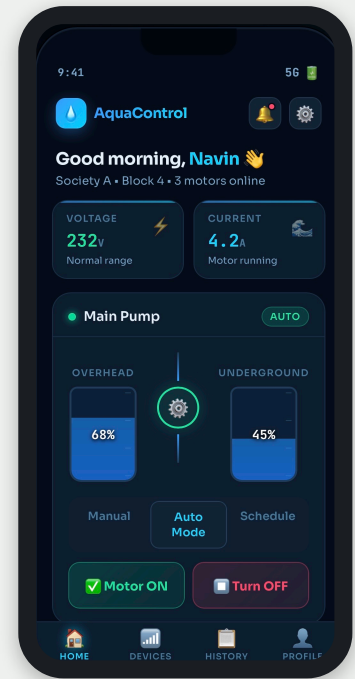
Tank-to-Tank Automatic Controller — App Controlled

SAME AUTOMATION · LIVE TANK LEVELS · REMOTE OVERRIDE

+ AQUACONTROL APP

Everything in the Tank-to-Tank Automatic Controller (Product 02), connected to the AquaControl app — so both tank levels are visible on a phone, with the option to step in remotely, not only from the panel.

- **Live tank levels** — overhead and underground tank percentage, updated on your phone.
- **Auto / Manual switch from the app** — change modes without a trip to the panel.
- **Remote override** — turn the transfer pump ON or OFF directly from the app.
- **Offline alert** — notified if the controller stops responding.
- **Multi-user access** — add household or society members with their own access to the device.



How-it-works logic, level control and technical specifications are unchanged from Product 02 (page 05) — the app adds remote visibility and control on top of the same controller.

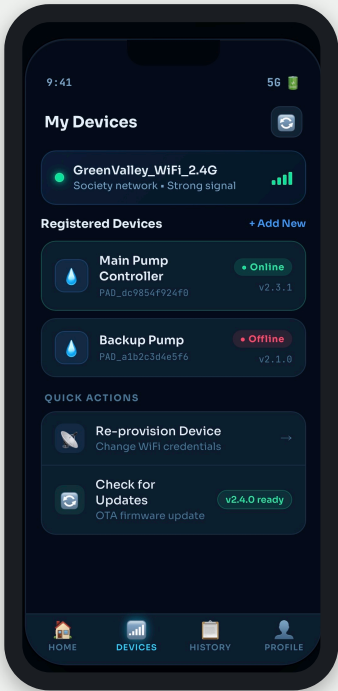
Borewell Pump Control & Protection Panel — App Controlled

SAME TIMER AUTOMATION · REMOTE STATUS · ACTIVITY LOG

+ AQUACONTROL APP

Everything in the Borewell Pump Control & Protection Panel (Product 03), connected to the AquaControl app — so pump status can be checked and overridden remotely, without a trip to the borewell.

- **Remote status check** — see whether the pump is running and whether the panel is online, from anywhere.
- **Remote Start / Stop** — override the timer schedule directly from the app when needed.
- **Offline alert** — notified the moment the panel stops responding — useful for remote borewell sites with no one nearby to notice.
- **Activity log** — see when the pump ran, for how long, and who triggered a manual change.
- **Multi-user access** — add members with their own access to the device.



Timer control, dry-run logic and technical specifications are unchanged from Product 03 (page 07) — the app adds remote visibility and control on top of the same panel.

Anywhere a pump needs watching



Farms & Irrigation

Borewell & field pumps — dry-run is the single most common way a farm motor dies. This is the failure mode FourFold exists to stop.



Buildings & Societies

Overhead-to-underground tank transfer for apartments and commercial buildings — automatic, with a manual panel for the caretaker.



Farmhouses

Independent homes on their own borewell — set it up once, then check on it from the app when you're not on-site.